



Literature Review & Research Ethics

PPT PRESENTED BY

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ROLL NO : 257Y1D5805

YEAR : I Year I Semester

ACADEMIC YEAR : 2025 - 2026

Effective Literature Studies

A Literature Review is a systematic search, analysis, and synthesis of published academic research. It provides background context, tracks prior studies, and identifies gaps for your project.

Objectives of Literature Review

- **Map the Domain:** Understand the historical development and current state of research in the field.
- **Identify Gaps:** Locate contradictions, limitations, or unaddressed areas in prior studies.
- **Avoid Duplication:** Verify that your proposed solution has not already been published.
- **Find Methodologies:** Learn from experimental setups, database testbeds, and statistical techniques used by others.

Systematic Review Approach

- **Keyword Search:** Formulate queries for databases like IEEE Xplore, ACM Digital Library, Google Scholar.
- **Filtering & Screening:** Screen papers by title, abstract, and then select candidate papers for full reading.
- **Synthesis vs. Summary:** Avoid simply listing papers. Group them by themes, compare their approaches, and outline trends in a structured synthesis.

Plagiarism in Research

Plagiarism is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit:

Forms of Plagiarism

- **Direct Plagiarism:** Word-for-word copy of text from a source without quotation marks or citations.
- **Paraphrased Plagiarism:** Copying ideas and rewriting sentences while keeping the original structure, without citation.
- **Self-Plagiarism (Duplicate Publication):** Re-using your own previously published code, text, or results in a new paper without citing the prior work.
- **Mosaic Plagiarism:** Patching together text from various sources, changing only a few words.

Prevention & Detection

- **Proper Citation:** Always cite the source when referencing ideas, data, diagrams, or quotes.
- **Use Quotation Marks:** Enclose exact copied phrases in quotes.
- **Paraphrase properly:** Synthesize ideas in your own words and write original sentences.
- **Checker Tools:** Run manuscripts through checker tools (e.g. Turnitin, iThenticate) before submission. Academic journals reject papers exceeding strict similarity limits (typically 10-15%).

Research Ethics & Integrity

Ethical guidelines protect the integrity of the scientific record and researchers:

Scientific Misconduct

- **Data Fabrication:** Making up fake experimental data or results and recording them as real.
- **Data Falsification:** Manipulating research equipment, omitting data points, or altering results to support a desired conclusion.
- **Selective Reporting:** Reporting only successful runs while omitting failed attempts or outliers.
- **Conflict of Interest:** Failing to disclose financial backing or relationships that could bias findings.

Publication Standards

- **Authorship Ethics:** Only include authors who made significant contributions to the design, execution, or writing. Avoid 'gift' or 'ghost' authors.
- **Duplicate Submission:** Submitting the same paper to multiple journals simultaneously is strictly prohibited.
- **Peer Review Integrity:** Reviewers must evaluate papers objectively, keep content confidential, and declare conflicts of interest.

Applications in Academic Publishing

Ethical guidelines and literature review standards are checked throughout the publishing cycle:

Journal Peer Reviewing

Submitted papers undergo rigorous peer review. Reviewers check the literature review section to ensure the researcher has mapped the field correctly and cited the latest advances.

University Thesis Defense

Graduate schools run thesis submissions through plagiarism checkers. Students must pass validation checks (similarity thresholds) to graduate.

Retraction of Publications

Journals retract published papers if fabrication, falsification, or plagiarism is exposed, permanently damaging the author's reputation and funding.

Open-Science Telemetry

Modern research sites require uploading raw data and code alongside the manuscript, enabling public replication and verifying findings.

THANK YOU

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